TEACHING WALLAH

Teaching Exam

Subject – Computer Science

Topic – Digital Logic

Lecture No.- 4

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Topics to be Covered

Topic

1:- Binary Arithmetic
2:- Binary Addition
3:- Binary Subtraction
4:- Binary subtraction using 1's complement
5:- Binary Subtraction using 2's complement
6:- Binary Multiplication
7:- Binary Division

0

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Binary arithmetic is one of the fundamental concepts in the field of digital electronics and computer engineering. It is basically the mathematics of binary numbers allow to perform various arithmetic operations on binary numbers. We know that the binary number system has two digits, i.e., 0 and 1 which are used to represent the ON or OFF states of the digital systems. Hence, binary arithmetic forms the foundation of the digital computing.

In this chapter, we will discuss the following four main binary arithmetic operations −

1:-Binary Addition

2:-Binary Subtraction

3:-Binary Multiplication

4:-Binary Division

What is Binary Addition

The binary addition operation works similarly to the base 10 decimal system, except that it is a base 2 system. The binary system consists of only two digits, 1 and 0. Most of the functionalities of the computer system use the binary number system. The binary code uses the digits 1's and 0's to make certain processes turn off or on. The process of the addition operation is very familiar to the decimal system by adjusting to the base 2.

Before attempting the binary addition process, we should have complete knowledge of how the place works in the binary number system. Because most of the modern digital computers and electronic circuits perform the binary operation by representing each bit as a voltage signal. The bit 0 represents the "OFF" state, and the bit 1 represents the "ON" state.

Rules of Binary Addition

Binary addition is much easier than the decimal addition when you remember the following tricks or rules. Using these rules, any binary number can be easily added. The four rules of binary addition are:

- 0 + 0 = 0
- 0 + 1 = 1
- 1 + 0 = 1
- 1 + 1 =10

$\boxed{x+y}$ - 2

$\to \boxed{p-2}$

$\boxed{(1\ 0)_2} \to (2)_{10}$

$2^1 \quad 2^0$

1:- 101010111 9bit
+ 101000011 9bit

1010011010

10bit

2:- 10000111010
+ 01010101010

11011100100 11bit

3:- 10101010
+ 10101010

101010100

1010.1101
+1100.0100

10111.0001

Binary Subtraction is a mathematical operation and is one of the four fundamental operations of binary numbers. Binary subtraction is similar to decimal subtraction but the rules used as somewhat different.

- Binary subtraction is a fundamental idea in binary operations.
- There are two components in binary subtraction: 0 and 1.

Binary Subtraction is a mathematical operation and is one of the four fundamental operations of binary numbers. Binary subtraction is similar to decimal subtraction but the rules used as somewhat different.

Borrow 1

$(10)_2 \to 2$

$(1)_2 \to 1$

$\cancel{1}(10)_2 \to (2)_{10}$

$-\ 0\ 1 \to (1)_{10}$

1

- Binary subtraction is a fundamental idea in binary operations.
- There are two components in binary subtraction: **0** and **1**.

$0 - 1 = 1$

$2 \leftarrow (10)_2$

$1 \leftarrow (1)_2$

X	Y	X-Y
0	0	0
0 $(10)_2$	1 $(1)_2$	1 (Borrow 1) $(1)_2$
1	0	1
1	1	0

Topic : METHOD OF BINARY SUBTRACTION

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Method 1: Subtraction of Binary Numbers without Borrowing

Step 1:-In binary, the number 8 is represented as $(1000)_2$ and the number 25 is represented as $(11001)_2$. Now subtract $(1000)_2$ from $(11001)_2$.

$$\begin{array}{r} 11001 \\ -\,01000 \\ \hline 10001 \end{array}$$

Step 2: To subtract the numbers, use binary subtraction principles.

Let us begin this subtraction by subtracting the integers on the right and working our way up to the next **higher-order** digit.

To begin, subtract (1-0). This equals 1.

Likewise, we proceed to the next higher-order digit and subtract (0-0), which is **equal** to 0.

Again, subtract (0-0), which is equals to 0. Then, we subtract (1-1), the outcome is 0

As a result, the difference is 10001_2.

$$\begin{array}{r} 11001 \\ -\,1000 \\ \hline 10001 \end{array}$$

Method 1: Subtraction of Binary Numbers with Borrowing

Step 1:- In binary, the number 12 is represented as $(1100)_2$ and the number 26 is represented as $(11010)_2$. Now subtract $(1100)_2$ from $(11010)_2$. Put the numbers in the order given below

$$\begin{array}{r} 11010 \\ -\ 1100 \\ \hline \\ \hline \end{array}$$

Step 2: To subtract the numbers, use binary subtraction principles.
Start the subtraction from the rightmost digit and move toward the higher-order digits.
Subtract (0 - 0) → Result: 0
Next, subtract (1 - 0) → Result: 1
Then, subtract (0 - 1):
Borrow 1 from the next higher order digit.
Result of (0 - 1) after borrowing → 1
Next, subtract (1 - 1) → Result: 0
Finally, subtract 0 (from the previous result) - 1 (borrowed) → Result: 1
Final binary difference: **$(1110)_2$**

Subtraction
1's

$\boxed{1\ 1\ 0\ 1\ 0}$ → Minued
$-\ \boxed{1\ 1\ 0\ 0}$ → Subtrahend

$$\begin{array}{r} 11010 \\ -\ 1100 \\ \hline 1110 \\ \hline \end{array}$$

Rule (Step)

1. Take 1's complement to Subtrahend
2. Add Minuend and 1's complement of Subtrahend
3. If Extra Carry not come take 1's complement of Result than your final Result will come
4. If Extra carry comes, Ignore extra carry but Add this Extra carry to the Result than final Result will come

```
  1 1 0 0 1      → Minuend
- 0 0 1 0 0      → Subtrahend
  1 1 0 1 1      → 1's complement of Subtrahend
```

```
  1 1 0 0 1
- 0 0 1 0 0
```

Extra Carry

```
  1 1   0   1 1      (carries)
    1 1 0 0 1
  + 1 1 0 1 1
  -------------
Ignore  1 | 1 0 1 0 0   ---- Result
              +1
  -------------
    1 0 1 0 1   → final Result
```

final Result

1 0 1 0 1

Binary Subtraction Using 1's Complement

- The number **0** represents the positive sign
- The number **1** represents the negative sign

Procedures for Binary Subtraction by 1's Complement

- Write the 1's complement of the subtrahend
- Then add the 1's complement subtrahend with the minuend
- If the result has a carryover, then add that carry over in the least significant bit
- If there is no carryover, then take the 1's complement of the resultant, and it is negative.

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Ex-1 :- $(110101)_2 - (100101)_2$

subtrahend

=> Take 1's Complement of Subtrahend

(011010)

=> Add this 1's complement to with Minuend means

$$\begin{array}{r} 110101 \\ +\ 011010 \\ \hline \end{array}$$

~~1 001111 if carry 1 comes~~ Discard it and carry 1 to add

Remaining output .

Extra carry

Ignore

1 1 1

$\boxed{1}$ 0 0 1 1 1 1

+ 1

X 0 1 0 0 0 0

~~001111~~

~~+ 1~~

~~010000~~ result

Ex-1 :- $(101011)_2 - (111001)_2$

subtrahend

=> Take 1's Complement of Subtrahend

(000110)

=> Add this 1's complement to with Minuend means

```
   1 0 1 0 1 1
+  0 0 0 1 1 0
--------------
   1 1 0 0 0 1
```

~~1 1 0 0 0 1 if carry 1 not comes then takes 1's complement again~~

1's complement

~~0 0 1 1 1 0 output result~~

No Extra Carry $\boxed{\times}$ 1 1 0 0 0 1 Result

Take 1's Complement

$\boxed{0\,0\,1\,1\,1\,0}$ → final Result

Binary Subtraction Using 2's Complement

Binary subtraction with 2's complement entails adding the complement of the subtrahend (the number being subtracted) to the minuend (the amount being removed). Here's an illustration:

Let's subtract $(100010)_2$ from $(110110)_2$. In this case, the binary equivalent of 34 is $(100010)_2$ whereas the binary equivalent of 54 is $(110110)_2$.

$(1994.125)_8$

Step 1: First identify the Minuend and Subtrahend. In thi,**this**s Minuend is $(110110)_2$ and subtrahend is $(100010)_2$.

Step 2: Find the 1's complement of the subtrahend. The subtrahend is $(100010)_2$ and after 1's complement it is $(011101)_2$. Now add 1 to the 1's complement $(011101)_2 + 1 = (011110)_2$. So the 2's Complement is $(011110)_2$.

Step 3: Now add the minuend and the 2's complement of the subtrahend.

$$
\begin{array}{r l}
110110 & \text{(Minuend)} \\
+\,011110 & \text{2's Complement of Subtrahend} \\
\hline
1010100 &
\end{array}
$$

Binary Subtraction using 2'S Complement

Step-1 Take 1's Complement

Step-2 Add 1 to 1's Complement

add

1 0 0 0 0 0 0

0 0 0 1 0 1 1

1'S Complement 1 1 1 0 1 0 0

+1

1 1 1 0 1 0 1

2'S Complement Subtrahend

1 0 0 0 0 0 0

+1 1 1 0 1 0 1

0 1 1 0 1 0 1 Final Result

Binary Subtraction using 2's Complement

Rule (Step)

1. Take 2's Complement of Subtrahend

2. Add Minuend and 2's Complement of Subtrahend then final result will come.

$$
\begin{array}{r}
1\,0\,1\,1\,1 \quad \rightarrow \text{Minuend} \\
-\,1\,0\,1\,1 \quad \rightarrow \text{Subtrahend}
\end{array}
$$

$$
\begin{array}{r}
1\,0\,1\,1 \\
\downarrow 1\text{'s} \\
0\,1\,0\,0 \\
+\,1 \\
\hline
0\,1\,0\,1
\end{array}
\rightarrow \text{2's Complement of Subtrahend}
$$

$$
\begin{array}{r}
1\;\;1\;\;1\;\;\; \\
1\,0\,1\,1\,1 \\
+\,0\,1\,0\,1 \\
\hline
1\,1\,1\,0\,0
\end{array}
\quad \text{final Result}
$$

Binary Multiplication is performed in the same manner as decimal numbers are multiplied. However, there are some specific rules regarding the multiplication among the binary digits 0 and 1, which we need to follow while performing the division of Binary Multiplication. Binary Multiplication rules are shown in the Binary Multiplication Table below:

Table for Binary Multiplication Rule

Rules for Multiplication	Meaning
$0 \times 0 = 0$	If 0 (zero) is multiplied by another 0 (zero), then the result is 0 (zero).
$0 \times 1 = 0$	If 0 (zero) is multiplied by 1 (one), then the result is 0 (zero).
$1 \times 0 = 0$	If 1 (one) is multiplied by 0 (zero), then the result is 0 (zero).
$1 \times 1 = 1$	If 1 (one) is multiplied by another 1 (one), then the result is 1 (one).

How to do Binary Multiplication?

There are five key steps involved in Binary Multiplication:,

Step 1: Write the multiplicand and the multiplier one below the other, aligning the rightmost digits.

Step 2: Multiply the multiplicand by each digit of the multiplier, starting from the rightmost digit.

Step 3: Move to the next digit of the multiplier and multiply it by the multiplicand. Write the result below the multiplier after shifting it to the left by one position.

Step 4: Repeat this process for each digit of the multiplier, shifting the result to the left by one more position each time.

Step 5: Add all the results using binary addition rules. The final sum is the product of the two binary numbers.

You can also use the Binary Multiplication Calculator to easily calculate the multiplication of two binary numbers.

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You can also use the Binary Multiplication Calculator to easily calculate the multiplication of two binary numbers.

Question 1: $(1010)_2 \times (101)_2$

Step 1: Write the multiplicand $(1010)_2$ and the multiplier $(101)_2$ one below the other, aligning the rightmost digits.

Step 2: Multiply the multiplicand by each digit of the multiplier, starting from the rightmost digit. The rightmost digit of the multiplier is 1, so we write down the multiplicand as it is.

Step 3: Move to the next digit of the multiplier and multiply it by the multiplicand. The next digit of the multiplier is 0, so we write down 0 after shifting it to the left by one position.

Step 4: Move to the next digit of the multiplier and multiply it by the multiplicand. The next digit of the multiplier is 1, so we write down the multiplicand after shifting it to the left by two positions.

Step 5: Add all the results using binary addition rules. The final sum $(110010)_2$ is the product of the two binary numbers.

```
      1 0 1 0
    x   1 0 1
  -----------
      1 0 1 0
    0 0 0 0 *
+ 1 0 1 0 * *
  -----------
  1 1 0 0 1 0
```

$(1010)_2 \times (101)_2 = (110010)_2$

So, the product of $(1010)_2 \times (101)_2$ is $(110010)_2$

Topic : BINARY MULTIPLICATION

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Question 3: $(0.101)_2 \times (0.11)_2$

Step 1: *Write the multiplicand $(0.101)_2$ and the multiplier $(0.11)_2$ one below the other, aligning the rightmost digits.*

Step 2: *Multiply the multiplicand by each digit of the multiplier, starting from the rightmost digit. The rightmost digit of the multiplier is 1, so we write down the multiplicand as it is.*

Step 3: *Move to the next digit of the multiplier and multiply it by the multiplicand. The next digit of the multiplier is 1, so we write down the multiplicand after shifting it to the left by one position.*

Step 4: *Move to the next digit of the multiplier and multiply it by the multiplicand. The next digit of the multiplier is 0, so we write down 0 after shifting it to the left by two positions.*

Step 5: *Add all the results using binary addition rules. The final sum $(001111)_2$ is the product of the two binary numbers.*

Step 6: *Count the total number of digits (d) after the decimal point in both the multiplicands and the multiplier. Place the decimal point in the product after (d) digits from right side.*

```
     0 . 1 0 1
   x   0 . 1 1
   -----------
         1 0 1
       1 0 1 *
  + 0 0 0 0 * *
   -----------
   0 0 1 1 1 1
 = 0 . 0 1 1 1 1
```

$(0.101)_2 \times (0.11)_2 = (0.01111)_2$

So, the product of (0.101)2 × $(0.11)_2$ is $(0.01111)_2$

Binary division is a mathematical operation that involves dividing two binary numbers, which are numbers composed of only 0's and 1's. Binary division is similar to decimal division, except that the base of the number system is 2 instead of 10.

Rules for Binary Division

A ÷ B	Result
0 ÷ 0	∞
0 ÷ 1	0
1 ÷ 1	1
1 ÷ 0	∞

Example 1: $(1011011)_2 \div (101)_2$

We start by taking the first three digits of the dividend $(101)_2$ which is equal to the divisor.

Step 1: *Write 1 as the first digit of the quotient. Then, we subtract the divisor from the first part of the dividend and write down the remainder.*

Step 2: *Next, we bring down the next digit of the dividend (1). Now we have $(1)_2$ which is less than the divisor $(101)_2$. So, we write 0 in the quotient.*

Step 3: *Next, we bring down the next digit of the dividend (0). Now we have $(10)_2$ which is less than the divisor $(101)_2$. So, we write 0 in the quotient.*

Step 4: *Next, we bring down the next digit of the dividend (1). Now we have $(101)_2$ which is equal to the divisor $(101)_2$. So, we write 1 in the quotient. We subtract the divisor from the current part of the dividend and write down the remainder.*

Step 5: *Finally, we bring down the last digit of the dividend (1). Now we have $(1)_2$ which is less than the divisor $(101)_2$. So, we write 0 in the quotient and 1 as the remainder.*

If This Value is always Greater or Equal then it gives quotient as 1
If It is less then take quotient as 0 till it comes to equal or greater

```
        10010
     ________
101 ) 1011011
     - 101
       ----
        0101
      -  101
        ----
          01
```

$(1011011)_2 \div (101)_2 = (10010)_2$

So, the quotient of $(1011011)_2 \div (101)_2$ is $(10010)_2$ and the remainder is $(1)_2$

Topic : BINARY DIVISON

EXAMPLE 2:- $(1010011.1010)_2 \div (100)_2$

Step 1: *Write 1 as the first digit of the quotient. Then, we subtract the divisor from the first part of the dividend and write down the remainder.*

Step 2: *Next, we bring down the next digit of the dividend (0). Now we have $(10)_2$ which is less than the divisor $(100)_2$. So, we write 0 in the quotient.*

Step 3: *Next, we bring down the next digit of the dividend (0). Now we have $(100)_2$ which is equal to the divisor $(100)_2$. So, we write 1 in the quotient. We subtract the divisor from the current part of the dividend and write down the remainder.*

Step 4: *Next, we bring down the next digit of the dividend (1). Now we have $(1)_2$ which is less than the divisor $(100)_2$. So, we write 0 in the quotient.*

Step 5: *Next, we bring down the next digit of the dividend (1). Now we have $(11)_2$ which is less than the divisor $(100)_2$. So, we write 0 in the quotient.*

Step 6: *Next, we bring down the next digit of the dividend (.). This indicates that we are now moving into the fractional part of the division. We continue the process as before.*

Step 7: *Next, we bring down the next digit of the dividend (1). Now we have $(111)_2$ which is greater than the divisor $(100)_2$. So, we write 1 in the quotient. We subtract the divisor from the current part of the dividend and write down the remainder.*

Step 8: *Next, we bring down the next digit of the dividend (0). Now we have $(110)_2$ which is greater than the divisor $(100)_2$. So, we write 1 in the quotient. We subtract the divisor from the current part of the dividend and write down the remainder.*

Step 9: *Next, we bring down the next digit of the dividend (1). Now we have $(101)_2$ which is equal to the divisor $(100)_2$. So, we write 1 in the quotient. We subtract the divisor from the current part of the dividend and write down the remainder.*

Step 10: *Finally, we bring down the last two digits of the dividend (0). Now we have $(10)_2$ which is less than the divisor $(100)_2$. So, we write it as the remainder.*

```
        10100.1110
    ______________
100 ) 1010011.1010
     -100
     ----
        100
      - 100
      -----
          111
        - 100
        -----
           110
         - 100
         -----
            101
          - 100
          -----
             10
```

$(1010011.1010)_2 \div (100)_2 = (10100.110)_2$

Q1. Divide $(10110)_2$ by $(10)_2$

Q2. Is $(10010101)_2$ is a multiple of $(11)_2$?

Q3. Divide $(11001110)_2$ by $(1001)_2$

Q4. Divide $(11110010)_2$ by $(1010)_2$

Q5. Divide $(11010)_2$ by $(101)_2$

Q6. $(10011001)_2 \div (1001)_2$

THANK YOU